MSC 201 **54855**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For an exponential distribution with mean 5 minutes, how does its median compare?	A. The median exceeds 5. B. The median does not exist. C. The median equals 5. D. The median is below 5. 10 pts
2. A recorded value has z=-1.5. What does that mean?	A. It has probability -1.5. B. It is above 98.5% of values automatically. C. It is 1.5 SD below the mean. D. It is 1.5 original units below zero. 10 pts
3. X is uniform from 0 to 10. Which event is more likely?	A. They are equally likely. B. X<6. C. X>6. D. X=6. 10 pts
4. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. Every item lasts exactly 8 hours. B. Every set totals exactly 80 hours. C. All items fail simultaneously at hour 80. D. It is an average total over repeated sets, not a guaranteed total. 10 pts
5. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. 0.5. B. 0.8849. C. 0.1151. D. -0.1151. 10 pts
6. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. Both standard deviations are zero. B. The second is larger because its mean is larger. C. The first is larger because it includes zero. D. They are equal. 10 pts
7. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It exceeds 1 automatically. B. It increases. C. It decreases. D. It stays unchanged. 10 pts

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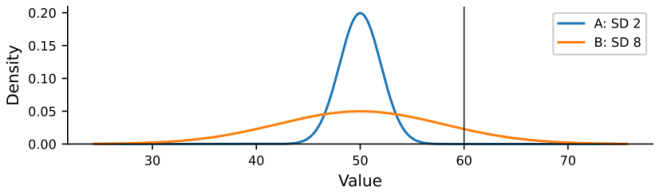
MSC 201 **54855**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Yes; rounding collects a positive-width interval of exact values. B. Only if the model becomes discrete before rounding. C. Only if the density equals zero everywhere. D. No; 4 seconds can never be recorded. 10 pts
9. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. They have equal upper tails. B. A. C. Both probabilities are exactly zero. D. B. 10 pts
10. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. A percentage of sample size. B. Probability units only. C. Mass units. D. No units because it is a z-score automatically. 10 pts

54855

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6.

D.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **57192**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Both must be uniform. B. Uniform first; normal second. C. Exponential first; normal second. D. Normal first; exponential second. 10 pts
2. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; total area may be any positive number. B. No; its total area is 1, and height is not probability. C. Yes; its probability is 2 everywhere. D. Yes; every density height must be at most 1. 10 pts
3. A normal variable has mean 40 and SD 5. What z-score corresponds to 50?	A. -2. B. 10. C. 0.2. D. 2. 10 pts
4. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The center shifts right. B. The total area rises above 1. C. It becomes lower as the curve spreads out. D. It becomes higher and narrower. 10 pts
5. P(X>2)=0.50 and P(X>6)=0.125 for an exponential wait. What is P(2<X<6)?	A. 0.625. B. 0.50. C. 0.375. D. 0.125. 10 pts
6. A continuous uniform model runs from 4 to 12. What is its mean?	A. 4. B. 8. C. 12. D. 16. 10 pts

Test continues on the other side of the sheet.

MSC 201 **57192**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. An exponential arrival rate doubles. What happens to mean waiting time?	A. It stays unchanged. B. It halves. C. It becomes zero. D. It doubles. 10 pts
8. For a bell-shaped normal model, about 68% lies within 1 SD. About how much lies outside that central interval?	A. 68%. B. 32%. C. 16% in total. D. 95%. 10 pts
9. For a normal model, where is the 10th percentile?	A. Below the mean. B. Above the mean. C. At the mean. D. At the largest possible value. 10 pts
10. Observed waits cluster near 0 and near 12, with few in the middle. What does that suggest about U(0,12)?	A. The uniform model is automatically proved. B. Its flat-density assumption deserves checking. C. A mean of 6 guarantees the uniform model. D. The range alone establishes the density. 10 pts

57192

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **50593**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 3/6. B. 9/12. C. 3/12. D. 6/12. 10 pts
2. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. Mean rises by 4; SD stays unchanged. B. Both rise by 4. C. SD rises by 4; mean stays unchanged. D. Both stay unchanged. 10 pts
3. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 44 to 56. B. 50 to 56. C. 48 to 52. D. 46 to 54. 10 pts
4. For a normal variable, a cutoff above the mean is reported to have upper-tail probability 0.80. What should be checked?	A. Whether probabilities can exceed 0.5 at all. B. Whether every normal curve has mean zero. C. Whether an upper tail must be negative. D. The tail direction; that upper tail must be below 0.5. 10 pts
5. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. 2. B. 3. C. -8. D. -2. 10 pts
6. The same process supplies arrivals per hour and time until next arrival. Which assignment is appropriate?	A. Uniform for both automatically. B. Poisson for the count; exponential for the wait. C. Exponential for the count; Poisson for the wait. D. Binomial for both automatically. 10 pts

Test continues on the other side of the sheet.

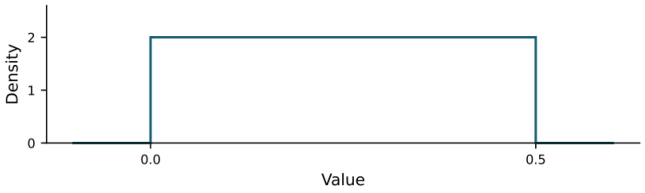
MSC 201 **50593**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; total area may be any positive number. B. Yes; every density height must be at most 1. C. No; its total area is 1, and height is not probability. D. Yes; its probability is 2 everywhere. 10 pts
8. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It falls by 3. B. It rises by 90%. C. It stays unchanged. D. It rises by 3. 10 pts
9. Two normal curves share SD but have different means. What is equal?	A. Their 90th-percentile values. B. Their probabilities below every fixed cutoff. C. Their centers. D. Their widths and shapes. 10 pts
10. An exponential model has mean wait 4 minutes. What is its SD?	A. 16 minutes. B. 0 minutes. C. 2 minutes. D. 4 minutes. 10 pts

50593

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **28088**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. They cluster near 6; 40 is the sample size. B. The model allows only the two values 6 and 40. C. The readings cluster near 40; 6 describes their spread. D. Every reading must equal 40. 10 pts
2. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Express rate and cutoff in compatible time units. B. Multiply both numbers by the sample size. C. Round the probability to an integer. D. Replace the cutoff with the mean. 10 pts
3. A continuous uniform model runs from 4 to 12. What is its mean?	A. 8. B. 4. C. 16. D. 12. 10 pts
4. Only 5% exceed cutoff k. Which percentile is k?	A. 100th. B. 50th. C. 5th. D. 95th. 10 pts
5. A fitted normal model predicts 2% above a cutoff, while a large observed dataset has 18% above it. What should be investigated?	A. Whether probabilities above 2% are impossible. B. Only the number of decimal places. C. Whether observations should be changed to fit the model. D. Whether the model fits the observed tail. 10 pts
6. A value equals its distribution's mean. What is its z-score?	A. 1. B. 0. C. The mean. D. The SD. 10 pts

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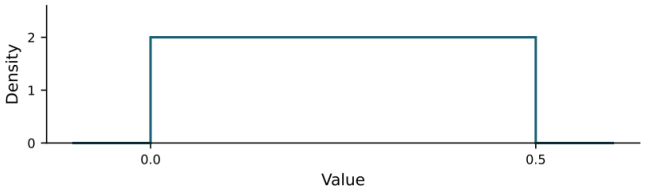
MSC 201 **28088**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem?  	A. No; its total area is 1, and height is not probability. B. Yes; its probability is 2 everywhere. C. No; total area may be any positive number. D. Yes; every density height must be at most 1. 10 pts
8. Observed waits cluster near 0 and near 12, with few in the middle. What does that suggest about U(0,12)?	A. A mean of 6 guarantees the uniform model. B. The range alone establishes the density. C. The uniform model is automatically proved. D. Its flat-density assumption deserves checking. 10 pts
9. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. 0.5. B. 0.8849. C. 0.1151. D. -0.1151. 10 pts
10. After an 11-minute wait, U(0,12) allows at most 1 more minute. An exponential model allows longer additional waits. What explains the difference?	A. Uniform waits have a finite endpoint; exponential waits do not. B. Exponential waits stop at their mean. C. Both models are memoryless. D. Uniform waits are always zero. 10 pts

28088

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10. A.

10 pts

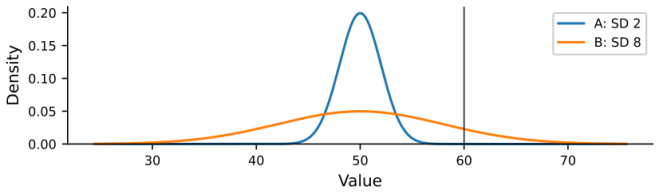
MSC 201 **40022**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. They are equally distant on opposite sides. B. Both lie above it. C. Both equal it. D. Q3 must be twice Q1. 10 pts
2. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They stay unchanged. B. They all become positive. C. They multiply by 100. D. They divide by 100. 10 pts
3. A proposed density dips below zero over an interval. What is wrong?	A. A density must be flat everywhere. B. Every valid density must be symmetric. C. Its maximum must equal 1. D. It would assign negative area to that interval. 10 pts
4. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. They have equal upper tails. B. B. C. Both probabilities are exactly zero. D. A. 10 pts
5. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. A. B. The one with the larger mean regardless of width. C. They have equal height. D. B. 10 pts
6. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They must occur as often as short waits. B. They are possible but less common than short waits. C. Every long wait has probability 1. D. They are impossible once the mean is exceeded. 10 pts

Test continues on the other side of the sheet.

MSC 201 **40022**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Which model assigns equal probability to all equal-length subintervals within its support?	A. A continuous uniform model. B. Every continuous model. C. An exponential model. D. A normal model. 10 pts
8. A normal model assigns 0.90 inside an accepted interval. What probability is outside it?	A. 1.90. B. 0.90. C. 0.10. D. 0.05 necessarily. 10 pts
9. A constant-rate process averages 0.5 requests per minute. What is the mean request count over 10 minutes?	A. 10. B. 20. C. 0.5. D. 5. 10 pts
10. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. 1 minute. B. 11 minutes. C. 12 minutes. D. There is no upper bound. 10 pts

40022

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. D.

10 pts

4. B.

10 pts

5. A.

10 pts

6. B.

10 pts

TEST PAGE 2

7. A.

10 pts

8. C.

10 pts

9. D.

10 pts

10. A.

10 pts

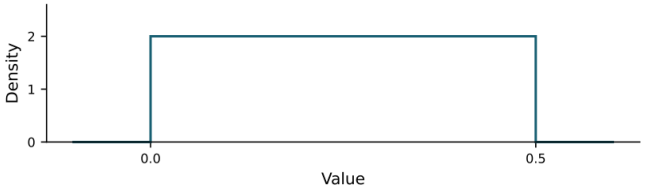
MSC 201 **65570**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. B. B. The one with the larger mean regardless of width. C. They have equal height. D. A. 10 pts
2. A normal variable has mean 40 and SD 5. What z-score corresponds to 50?	A. -2. B. 0.2. C. 10. D. 2. 10 pts
3. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It becomes 1. B. It doubles. C. It stays unchanged. D. It halves. 10 pts
4. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Recording waits in minutes. B. A constant rate across the whole period. C. Nonnegative waiting times. D. Allowing a long wait. 10 pts
5. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. Yes; every density height must be at most 1. B. Yes; its probability is 2 everywhere. C. No; its total area is 1, and height is not probability. D. No; total area may be any positive number. 10 pts
6. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. It assigns appreciable probability to impossible values. B. Its mean must be zero. C. Every duration must be negative. D. Normal distributions cannot model any positive variable. 10 pts

Test continues on the other side of the sheet.

MSC 201 **65570**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. For a normal variable, a cutoff above the mean is reported to have upper-tail probability 0.80. What should be checked?	A. The tail direction; that upper tail must be below 0.5. B. Whether every normal curve has mean zero. C. Whether an upper tail must be negative. D. Whether probabilities can exceed 0.5 at all. 10 pts
8. A supplied exponential model gives P(X>8)=0.20. What is P(X<=8)?	A. 0.40. B. 0.20. C. 0.80. D. 1.20. 10 pts
9. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Exponential first; normal second. B. Both must be uniform. C. Uniform first; normal second. D. Normal first; exponential second. 10 pts
10. Only 5% exceed cutoff k. Which percentile is k?	A. 95th. B. 100th. C. 50th. D. 5th. 10 pts

65570

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

C.

10 pts

6. A.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

C.

10 pts

9.

D.

10 pts

10. A.

10 pts

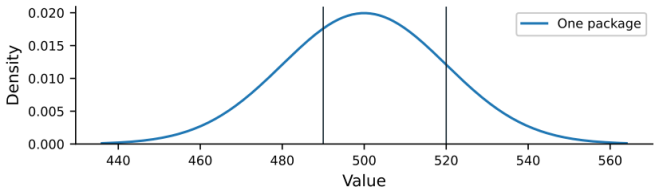
MSC 201 **40019**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.15%. B. 3%. C. 0.3%. D. 99.7%. 10 pts
2. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. They are equal. B. The second is larger because its mean is larger. C. The first is larger because it includes zero. D. Both standard deviations are zero. 10 pts
3. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. 3. B. -2. C. 2. D. -8. 10 pts
4. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Only the point 500. B. Everything above 520. C. Only the area between 490 and 520. D. Both outside tails. 10 pts
5. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The total area rises above 1. B. It becomes higher and narrower. C. The center shifts right. D. It becomes lower as the curve spreads out. 10 pts
6. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 10. B. At 1. C. At 5. D. At 9. 10 pts

Test continues on the other side of the sheet.

MSC 201 **40019**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They are impossible once the mean is exceeded. B. They must occur as often as short waits. C. Every long wait has probability 1. D. They are possible but less common than short waits. 10 pts
8. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. All items fail simultaneously at hour 80. B. It is an average total over repeated sets, not a guaranteed total. C. Every item lasts exactly 8 hours. D. Every set totals exactly 80 hours. 10 pts
9. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 45%. B. 10%. C. 5%. D. 90%. 10 pts
10. F(3)=0.20 and F(7)=0.80 for a continuous variable. What is P(3<X<7)?	A. 0.20. B. 0.60. C. 1.00. D. 0.80. 10 pts

40019

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

B.

10 pts

9.

C.

10 pts

10.

B.

10 pts

MSC 201 **54255**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is larger than P(X>8). B. It is 1/2. C. It is the density height at 12. D. It is zero. 10 pts
2. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. They are equally distant on opposite sides. B. Both lie above it. C. Both equal it. D. Q3 must be twice Q1. 10 pts
3. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. It becomes right-skewed. B. It shifts right without changing width. C. Its total area increases. D. It becomes wider. 10 pts
4. Observed waits cluster near 0 and near 12, with few in the middle. What does that suggest about U(0,12)?	A. The range alone establishes the density. B. The uniform model is automatically proved. C. A mean of 6 guarantees the uniform model. D. Its flat-density assumption deserves checking. 10 pts
5. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 0.2. B. 25. C. 0.5. D. 5. 10 pts
6. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 95%. B. 2.5%. C. 5%. D. 47.5%. 10 pts
7. A recorded value has z=-1.5. What does that mean?	A. It has probability -1.5. B. It is 1.5 SD below the mean. C. It is 1.5 original units below zero. D. It is above 98.5% of values automatically. 10 pts

Test continues on the other side of the sheet.

MSC 201 **54255**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The model with the larger SD. B. The model with the smaller SD. C. They must have equal tails. D. The one with more recorded observations regardless of SD. 10 pts
9. A rectangular density covers 0 to 4 and is zero elsewhere. What height makes total area 1?	A. 4. B. 1. C. 0.5. D. 0.25. 10 pts
10. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. A constant rate across the whole period. B. Nonnegative waiting times. C. Recording waits in minutes. D. Allowing a long wait. 10 pts

54255

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6.

B.

10 pts

7.

B.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

D.

10 pts

10. A.

10 pts

MSC 201 **62828**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Every duration must be negative. B. Its mean must be zero. C. Normal distributions cannot model any positive variable. D. It assigns appreciable probability to impossible values. 10 pts
2. A normal variable has mean 40 and SD 5. What z-score corresponds to 50?	A. -2. B. 0.2. C. 2. D. 10. 10 pts
3. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.3%. B. 0.15%. C. 99.7%. D. 3%. 10 pts
4. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 25. B. 0.2. C. 5. D. 0.5. 10 pts
5. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. No; the average has less spread. B. Yes; both are measured in grams. C. Yes; averages always vary more. D. No; averages have no units. 10 pts
6. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Flat over 4 to 12. B. Normal over all real numbers. C. Flat over 0 to 12 unchanged. D. A point at 4. 10 pts
7. Only 5% exceed cutoff k. Which percentile is k?	A. 100th. B. 5th. C. 50th. D. 95th. 10 pts

Test continues on the other side of the sheet.

MSC 201 **62828**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. X is uniform from 0 to 12. Which symmetric interval contains the middle 50%?	A. 0 to 6. B. 3 to 9. C. 5 to 7. D. 6 to 12. 10 pts
9. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. The same under the memoryless model. B. It must end immediately. C. It becomes uniform. D. Its expected remaining duration is the original mean minus 10. 10 pts
10. A proposed density dips below zero over an interval. What is wrong?	A. Its maximum must equal 1. B. Every valid density must be symmetric. C. A density must be flat everywhere. D. It would assign negative area to that interval. 10 pts

62828

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3. A.

10 pts

4.

B.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **96068**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 12.8. B. 101.28. C. 87.2. D. 112.8. 10 pts
2. A wait is uniform from 0 to 20 minutes. What is its 75th percentile?	A. 15 minutes. B. 20 minutes. C. 5 minutes. D. 75 minutes. 10 pts
3. For one normal model, P(X<40)=0.20 and P(X<60)=0.85. What is P(40<X<60)?	A. 0.20. B. 0.65. C. 1.05. D. 0.85. 10 pts
4. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It decreases. B. It increases. C. It stays unchanged. D. It exceeds 1 automatically. 10 pts
5. A nonnormal variable has a well-defined mean and SD. Can its z-score automatically be looked up as a standard-normal probability?	A. No; only normal variables can be standardized. B. Yes; a positive SD is sufficient. C. Yes; z-scores always imply normality. D. No; the normal shape assumption still matters. 10 pts
6. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. It becomes wider. B. It becomes right-skewed. C. Its total area increases. D. It shifts right without changing width. 10 pts

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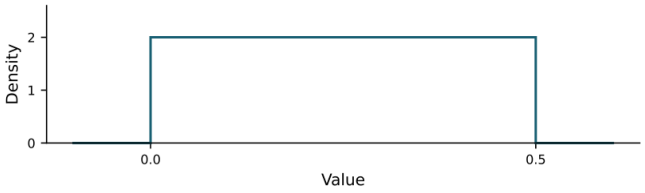
MSC 201 **96068**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. Yes; its probability is 2 everywhere. B. No; its total area is 1, and height is not probability. C. Yes; every density height must be at most 1. D. No; total area may be any positive number. 10 pts
8. A supplied exponential model gives P(X>8)=0.20. What is P(X<=8)?	A. 0.20. B. 0.40. C. 1.20. D. 0.80. 10 pts
9. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Flat over 0 to 12 unchanged. B. Normal over all real numbers. C. Flat over 4 to 12. D. A point at 4. 10 pts
10. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. All items fail simultaneously at hour 80. B. Every set totals exactly 80 hours. C. It is an average total over repeated sets, not a guaranteed total. D. Every item lasts exactly 8 hours. 10 pts

96068

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10.

C.

10 pts

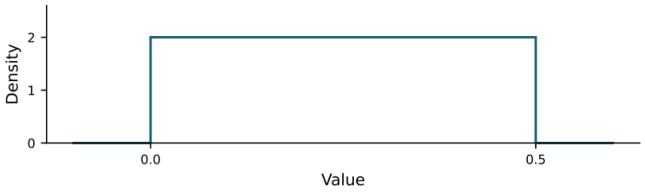
MSC 201 **69292**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; total area may be any positive number. B. No; its total area is 1, and height is not probability. C. Yes; its probability is 2 everywhere. D. Yes; every density height must be at most 1. 10 pts
2. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The center shifts right. B. It becomes lower as the curve spreads out. C. The total area rises above 1. D. It becomes higher and narrower. 10 pts
3. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 90%. B. 5%. C. 10%. D. 45%. 10 pts
4. An exponential arrival rate doubles. What happens to mean waiting time?	A. It halves. B. It stays unchanged. C. It becomes zero. D. It doubles. 10 pts
5. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. Every long wait has probability 1. B. They are possible but less common than short waits. C. They must occur as often as short waits. D. They are impossible once the mean is exceeded. 10 pts
6. $P(X>50)=0.50$ and $P(X>60)=0.10$. Given $X>50$, what is $P(X>60)$?	A. 0.40. B. 0.10. C. 0.60. D. 0.20. 10 pts

Test continues on the other side of the sheet.

MSC 201 **69292**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A recorded value has z=-1.5. What does that mean?	A. It is 1.5 original units below zero. B. It has probability -1.5. C. It is 1.5 SD below the mean. D. It is above 98.5% of values automatically. 10 pts
8. X is uniform from 2 to 10. To find P(0<X<5), which portion contributes area?	A. 2 to 5. B. 0 to 5. C. 0 to 2. D. 5 to 10. 10 pts
9. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. Both standard deviations are zero. B. They are equal. C. The first is larger because it includes zero. D. The second is larger because its mean is larger. 10 pts
10. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. 0.1151. B. -0.1151. C. 0.8849. D. 0.5. 10 pts

69292

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8. A.

10 pts

9.

B.

10 pts

10. A.

10 pts

MSC 201 **88735**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is 1/2. B. It is larger than P(X>8). C. It is the density height at 12. D. It is zero. 10 pts
2. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. No; exponential values must be negative. B. No; the arrival process and distributional assumptions still need support. C. Yes; every nonnegative variable is exponential. D. Yes; nonnegative variables have equal mean and SD. 10 pts
3. A continuous model gives F(6)=0.70. What does that say?	A. The density height at 6 is 0.70. B. 70% of modeled durations are at most 6. C. The mean is 6. D. 70% of durations equal exactly 6. 10 pts
4. A normal model has mean 20. What is P(X<20)?	A. The SD. B. 1. C. 0. D. 0.5. 10 pts
5. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. They are equal. B. The first is larger because it includes zero. C. The second is larger because its mean is larger. D. Both standard deviations are zero. 10 pts
6. P(X>50)=0.50 and P(X>60)=0.10. Given X>50, what is P(X>60)?	A. 0.20. B. 0.10. C. 0.60. D. 0.40. 10 pts
7. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It rises by 90%. B. It stays unchanged. C. It falls by 3. D. It rises by 3. 10 pts

Test continues on the other side of the sheet.

MSC 201 **88735**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. For one normal model, P(X<40)=0.20 and P(X<60)=0.85. What is P(40<X<60)?	A. 0.65. B. 0.20. C. 1.05. D. 0.85. 10 pts
9. A value equals its distribution's mean. What is its z-score?	A. 0. B. The mean. C. The SD. D. 1. 10 pts
10. Compare one exponential duration with the average of many independent durations. What remains the same?	A. The SD. B. The mean. C. The maximum possible duration becomes finite for the average. D. The entire density shape. 10 pts

88735

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **18482**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It rises by 3. B. It rises by 90%. C. It falls by 3. D. It stays unchanged. 10 pts
2. After an 11-minute wait, U(0,12) allows at most 1 more minute. An exponential model allows longer additional waits. What explains the difference?	A. Exponential waits stop at their mean. B. Both models are memoryless. C. Uniform waits are always zero. D. Uniform waits have a finite endpoint; exponential waits do not. 10 pts
3. For a bell-shaped normal model, about 68% lies within 1 SD. About how much lies outside that central interval?	A. 16% in total. B. 32%. C. 95%. D. 68%. 10 pts
4. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Flat over 4 to 12. B. Normal over all real numbers. C. Flat over 0 to 12 unchanged. D. A point at 4. 10 pts
5. A is 6 above its mean with SD 3; B is 8 above its mean with SD 8. Which is more extreme above its group center?	A. A. B. They are equally extreme. C. B. D. The larger raw difference always determines extremeness. 10 pts
6. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. It becomes wider. B. It becomes right-skewed. C. It shifts right without changing width. D. Its total area increases. 10 pts
7. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is the density height at 12. B. It is zero. C. It is 1/2. D. It is larger than P(X>8). 10 pts

Test continues on the other side of the sheet.

MSC 201 **18482**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Round the probability to an integer. B. Replace the cutoff with the mean. C. Express rate and cutoff in compatible time units. D. Multiply both numbers by the sample size. 10 pts
9. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 0 to 4 seconds. B. Only the exact point 4. C. 4 to 5 seconds. D. 3.5 to 4.5 seconds. 10 pts
10. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Normal first; exponential second. B. Uniform first; normal second. C. Exponential first; normal second. D. Both must be uniform. 10 pts

18482

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. B.

10 pts

4. A.

10 pts

5. A.

10 pts

6. C.

10 pts

7. B.

10 pts

TEST PAGE 2

8. C.

10 pts

9. D.

10 pts

10. A.

10 pts

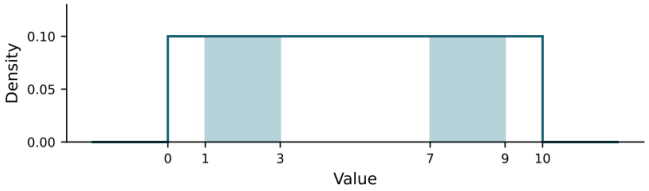
MSC 201 **27125**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. At most is larger by the density height. B. They are equal. C. They differ by 1/480. D. Below is always zero. 10 pts
2. X is uniform from 0 to 10. Compare intervals 1 to 3 and 7 to 9. 	A. Their probabilities depend on the sample median. B. They have equal probability. C. The first is more likely because it is closer to zero. D. The second is more likely because its values are larger. 10 pts
3. A constant-rate process averages 0.5 requests per minute. What is the mean request count over 10 minutes?	A. 10. B. 5. C. 0.5. D. 20. 10 pts
4. For an exponential distribution with mean 5 minutes, how does its median compare?	A. The median equals 5. B. The median is below 5. C. The median exceeds 5. D. The median does not exist. 10 pts
5. For a continuous duration, how do P(X<6) and P(X<=6) compare?	A. They differ by the density height at 6. B. The first is always zero. C. The second is always twice the first. D. They are equal. 10 pts
6. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 50 to 56. B. 44 to 56. C. 46 to 54. D. 48 to 52. 10 pts

Test continues on the other side of the sheet.

MSC 201 **27125**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. Both standard deviations are zero. B. They are equal. C. The second is larger because its mean is larger. D. The first is larger because it includes zero. 10 pts
8. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. A percentage of sample size. B. Probability units only. C. Mass units. D. No units because it is a z-score automatically. 10 pts
9. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. It becomes higher and narrower. B. The center shifts right. C. It becomes lower as the curve spreads out. D. The total area rises above 1. 10 pts
10. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. -2. B. -8. C. 3. D. 2. 10 pts

27125

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **67275**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Compare one U(0,12) wait with the average of 100 independent such waits. Which is less variable?	A. They have identical spread. B. The average must have zero spread. C. The average. D. The individual wait. 10 pts
2. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. They differ by 1/480. B. They are equal. C. Below is always zero. D. At most is larger by the density height. 10 pts
3. Normal models share SD 4 but have means 10 and 14. Which has more probability below 8?	A. The mean-14 model. B. Both have probability 0.5. C. They have equal probabilities. D. The mean-10 model. 10 pts
4. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The one with more recorded observations regardless of SD. B. They must have equal tails. C. The model with the smaller SD. D. The model with the larger SD. 10 pts
5. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 5. B. At 1. C. At 10. D. At 9. 10 pts
6. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. A percentage of sample size. B. Mass units. C. No units because it is a z-score automatically. D. Probability units only. 10 pts
7. An exponential density extends far right with diminishing height. What does that say about very long waits?	A. They are possible but less common than short waits. B. They are impossible once the mean is exceeded. C. Every long wait has probability 1. D. They must occur as often as short waits. 10 pts

Test continues on the other side of the sheet.

MSC 201 **67275**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It becomes uniform. B. It must end immediately. C. Its expected remaining duration is the original mean minus 10. D. The same under the memoryless model. 10 pts
9. Mean is 30 and SD is 4. Which value has z=1.5?	A. 36. B. 26. C. 45. D. 31.5. 10 pts
10. A rectangular density covers 0 to 4 and is zero elsewhere. What height makes total area 1?	A. 4. B. 0.25. C. 1. D. 0.5. 10 pts

67275

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

B.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10.

B.

10 pts

MSC 201 **80936**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They all become positive. B. They multiply by 100. C. They divide by 100. D. They stay unchanged. 10 pts
2. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. They have equal height. B. A. C. The one with the larger mean regardless of width. D. B. 10 pts
3. Which model assigns equal probability to all equal-length subintervals within its support?	A. A continuous uniform model. B. An exponential model. C. A normal model. D. Every continuous model. 10 pts
4. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 47.5%. B. 5%. C. 95%. D. 2.5%. 10 pts
5. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. It becomes right-skewed. B. Its total area increases. C. It becomes wider. D. It shifts right without changing width. 10 pts
6. Someone reports F(4)=0.6 and F(8)=0.4 for one variable. Why is this impossible?	A. Every cutoff must have the same probability. B. Cumulative functions must always equal 0.5. C. A cumulative probability cannot decrease as the cutoff rises. D. Both numbers are below 1. 10 pts
7. How does no arrivals in the next 5 minutes relate to waiting more than 5 minutes for the next arrival?	A. They are complements. B. The first requires exactly one arrival. C. They describe the same event, ignoring a zero-probability boundary. D. They are independent unrelated events. 10 pts

Test continues on the other side of the sheet.

MSC 201 **80936**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. They are equal. B. Both standard deviations are zero. C. The second is larger because its mean is larger. D. The first is larger because it includes zero. 10 pts
9. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 90%. B. 10%. C. 5%. D. 45%. 10 pts
10. An exponential model has mean wait 4 minutes. What is its SD?	A. 4 minutes. B. 2 minutes. C. 16 minutes. D. 0 minutes. 10 pts

80936

TEST PAGE 1

1.

D.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **48660**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Only 5% exceed cutoff k. Which percentile is k?	A. 95th. B. 5th. C. 100th. D. 50th. 10 pts
2. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Round the probability to an integer. B. Express rate and cutoff in compatible time units. C. Multiply both numbers by the sample size. D. Replace the cutoff with the mean. 10 pts
3. A proposed density dips below zero over an interval. What is wrong?	A. A density must be flat everywhere. B. It would assign negative area to that interval. C. Every valid density must be symmetric. D. Its maximum must equal 1. 10 pts
4. P(X>50)=0.50 and P(X>60)=0.10. Given X>50, what is P(X>60)?	A. 0.20. B. 0.40. C. 0.60. D. 0.10. 10 pts
5. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Normal over all real numbers. B. Flat over 4 to 12. C. Flat over 0 to 12 unchanged. D. A point at 4. 10 pts
6. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. 3. B. 2. C. -8. D. -2. 10 pts
7. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 3/12. B. 9/12. C. 6/12. D. 3/6. 10 pts

Test continues on the other side of the sheet.

MSC 201 **48660**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. All items fail simultaneously at hour 80. B. It is an average total over repeated sets, not a guaranteed total. C. Every set totals exactly 80 hours. D. Every item lasts exactly 8 hours. 10 pts
9. For a normal variable, a cutoff above the mean is reported to have upper-tail probability 0.80. What should be checked?	A. Whether every normal curve has mean zero. B. Whether an upper tail must be negative. C. The tail direction; that upper tail must be below 0.5. D. Whether probabilities can exceed 0.5 at all. 10 pts
10. For a standard normal variable, compare $P(Z<-1.4)$ and $P(Z>1.4)$.	A. The left tail is larger. B. They are equal. C. The right tail is larger. D. Both equal 0.5. 10 pts

48660

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. A.

10 pts

5. B.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. B.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **35554**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. F(3)=0.20 and F(7)=0.80 for a continuous variable. What is P(3<X<7)?	A. 0.60. B. 0.80. C. 1.00. D. 0.20. 10 pts
2. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Normal over all real numbers. B. Flat over 4 to 12. C. Flat over 0 to 12 unchanged. D. A point at 4. 10 pts
3. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. Both equal it. B. They are equally distant on opposite sides. C. Q3 must be twice Q1. D. Both lie above it. 10 pts
4. A wait is uniform from 0 to 20 minutes. What is its 75th percentile?	A. 15 minutes. B. 75 minutes. C. 5 minutes. D. 20 minutes. 10 pts
5. A nonnormal variable has a well-defined mean and SD. Can its z-score automatically be looked up as a standard-normal probability?	A. No; only normal variables can be standardized. B. Yes; a positive SD is sufficient. C. No; the normal shape assumption still matters. D. Yes; z-scores always imply normality. 10 pts
6. Requests follow a constant-rate Poisson process at 4 per hour. What is the mean waiting time to the next request?	A. 60 minutes. B. 15 minutes. C. 4 minutes. D. 240 minutes. 10 pts
7. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. At most is larger by the density height. B. They are equal. C. They differ by 1/480. D. Below is always zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **35554**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It must end immediately. B. The same under the memoryless model. C. Its expected remaining duration is the original mean minus 10. D. It becomes uniform. 10 pts
9. A normal model's mean increases, with SD fixed. At a fixed deadline, what happens to the probability of finishing before it?	A. It increases. B. It stays unchanged. C. It decreases. D. It becomes exactly 0.5. 10 pts
10. A normal model has mean 20. What is P(X<20)?	A. 0. B. 0.5. C. 1. D. The SD. 10 pts

35554

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. A.

10 pts

5. C.

10 pts

6. B.

10 pts

7. B.

10 pts

TEST PAGE 2

8. B.

10 pts

9. C.

10 pts

10. B.

10 pts

MSC 201 **78955**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A is 6 above its mean with SD 3; B is 8 above its mean with SD 8. Which is more extreme above its group center?	A. A. B. They are equally extreme. C. The larger raw difference always determines extremeness. D. B. 10 pts
2. Requests follow a constant-rate Poisson process at 4 per hour. What is the mean waiting time to the next request?	A. 60 minutes. B. 4 minutes. C. 240 minutes. D. 15 minutes. 10 pts
3. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It stays unchanged. B. It halves. C. It becomes 1. D. It doubles. 10 pts
4. An exponential arrival rate doubles. What happens to mean waiting time?	A. It stays unchanged. B. It becomes zero. C. It doubles. D. It halves. 10 pts
5. For a continuous duration, how do P(X<6) and P(X<=6) compare?	A. The second is always twice the first. B. They differ by the density height at 6. C. The first is always zero. D. They are equal. 10 pts
6. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It rises by 90%. B. It stays unchanged. C. It falls by 3. D. It rises by 3. 10 pts
7. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. They are equal. B. They differ by 1/480. C. Below is always zero. D. At most is larger by the density height. 10 pts

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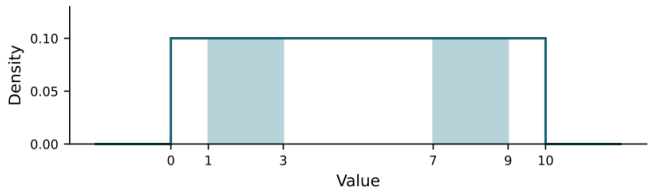
MSC 201 **78955**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Two normal curves share SD but have different means. What is equal?	A. Their 90th-percentile values. B. Their widths and shapes. C. Their probabilities below every fixed cutoff. D. Their centers. 10 pts
9. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. They must have equal tails. B. The model with the larger SD. C. The model with the smaller SD. D. The one with more recorded observations regardless of SD. 10 pts
10. X is uniform from 0 to 10. Compare intervals 1 to 3 and 7 to 9. 	A. The first is more likely because it is closer to zero. B. They have equal probability. C. The second is more likely because its values are larger. D. Their probabilities depend on the sample median. 10 pts

78955

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. B.

10 pts

4. D.

10 pts

5. D.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. B.

10 pts

9. B.

10 pts

10. B.

10 pts

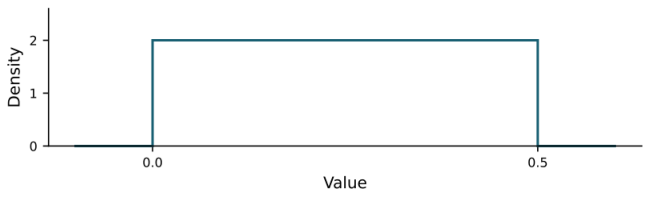
MSC 201 **47135**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Both must be uniform. B. Exponential first; normal second. C. Normal first; exponential second. D. Uniform first; normal second. 10 pts
2. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 5. B. At 9. C. At 1. D. At 10. 10 pts
3. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It becomes uniform. B. The same under the memoryless model. C. Its expected remaining duration is the original mean minus 10. D. It must end immediately. 10 pts
4. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. Yes; nonnegative variables have equal mean and SD. B. Yes; every nonnegative variable is exponential. C. No; exponential values must be negative. D. No; the arrival process and distributional assumptions still need support. 10 pts
5. For a normal model, where is the 10th percentile?	A. Above the mean. B. Below the mean. C. At the mean. D. At the largest possible value. 10 pts
6. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. Yes; every density height must be at most 1. B. Yes; its probability is 2 everywhere. C. No; total area may be any positive number. D. No; its total area is 1, and height is not probability. 10 pts

Test continues on the other side of the sheet.

MSC 201 **47135**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. At most is larger by the density height. B. Below is always zero. C. They differ by 1/480. D. They are equal. 10 pts
8. A student treats z=2 as a probability of 2. What is wrong?	A. A normal model cannot have positive z-scores. B. A z-score can never exceed 1. C. A z-score must be a percentage. D. A standardized distance is not an area probability. 10 pts
9. Observed waits cluster near 0 and near 12, with few in the middle. What does that suggest about U(0,12)?	A. A mean of 6 guarantees the uniform model. B. The range alone establishes the density. C. Its flat-density assumption deserves checking. D. The uniform model is automatically proved. 10 pts
10. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. Its total area increases. B. It becomes wider. C. It becomes right-skewed. D. It shifts right without changing width. 10 pts

47135

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

B.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

C.

10 pts

10.

D.

10 pts